名古屋大学大学院工学研究科博士課程（前期課程）

機械システム工学専攻
マイクロ・ナノ機械理工学専攻
航空宇宙工学専攻

語学試験問題

（全 6 頁，表紙を含む）

令和 7 年 11 月 19 日（水曜日）
10:00～11:40 (100 分)

注意

- すべての問題に解答せよ.
- 受験番号を答案用紙の各ページの所定欄に書くこと.
- 問題ごとに解答を別の答案用紙に記入し、左上に問題番号を書くこと.

Graduate School of Engineering
Nagoya University (Master's and Doctoral Programs)

Department of Mechanical Systems Engineering
Department of Micro-Nano Mechanical Science and Engineering
Department of Aerospace Engineering

Language Test

Number of pages (including this page): 6

Wednesday, November 19, 2025
10:00～11:40 (100 min.)

Notes

- Answer all the problems.
- Write your application number in the designated place at the top of each answer sheet.
- Use one answer sheet for each problem, and write its problem number on the top-left corner of the answer sheet.

Problem I

Read the following passage and answer the questions (1) - (7) in English.

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(Extracted from MIT News, The Climate and Sustainability Implications of Generative AI, Jan. 17, 2025)

(1) Choose the most appropriate meaning of (a) staggering.

a) surprising and enormous

b) decreasing and slow

c) steady and balanced

d) temporary and unstable

(2) Choose the most appropriate meaning of (b) strain.

a) make stronger

b) damage or put pressure on

c) clean or purify

d) connect together

(3) For the sentence (c), choose the appropriate words for (A) and (B).

i) Choices for (A):

a) a small system

b) an industrial

c) a system

d) a technological

ii) Choices for (B):

a) emotions

b) actions

c) predictions

d) technologies

(4) Describe the two main environmental factors affected by generative AI development and operation using expressions from the first paragraph of the passage.

(5) For the sentence (d), arrange the words in the correct order. Note that there may be extra word(s) that is not needed in the sentence.

(6) In the sentences (1) and (2), calculate the approximate annual growth rate (%) of electricity consumption assuming an exponential increase with the same rate each year.

(a) $(0.438)^{1/4} \sim 0.111$ (11.1%)
(b) $(2.283)^{1/2} \sim 1.511$ (151.1%)
(c) $(2.283)^{1/4} - 1 \sim 0.229$ (22.9%)
(d) $(2.283)^{1/2} - 1 \sim 0.511$ (51.1%)

(7) Summarize the passage within 100 words.

Problem II

1. Translate the following sentences into Japanese.
   (1) The efficiency of a solar cell depends largely on the materials used and the way they are structured at the microscopic level.
   (2) Recent advances in quantum computing have demonstrated that certain problems, previously considered to be hard to deal with, can be solved extraordinarily faster than by the classical computing.
   (3) When the temperature increases, the resistance of most metals also increases due to the scattering of electrons.
   (4) The experiment was repeated three times to ensure the reliability and reproducibility of the results.
   (5) Artificial intelligence has made it possible to detect patterns in large datasets that would be impossible for humans to analyze manually.

2. Translate the following sentences into English.
   (1) 非線形方程式系を解析的に解くことは困難であるため，数値的手法が広く用いられている．
   (2) このロボットは，センサーから得られたデータをもとに自律的に動作する．
   (3) 実験データとシミュレーション結果の整合性を確認するために，追加の検証が必要である．
   (4) データを正確に測定するためには，高精度の装置が必要だ．
   (5) これらの人工衛星は地球の表面を観測し，気候変動の研究に利用されている．

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機械システム工学専攻
マイクロ・ナノ機械理工学専攻
航空宇宙工学専攻

専門科目試験問題

（全 4 頁，表紙を含む）

令和 7 年 11 月 19 日（水曜日）

13:00～15:30 (150 分)

注意

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問題 I

1. 次を計算せよ．

(1) $\dfrac{d}{dx}\left(x^{x\sin x}\right)$

(2) $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x\sin x}$

2. 次の微分方程式の一般解を求めよ．

(1) $e^{2x-y}+e^{x+y}\dfrac{dy}{dx}=0$

(2) $\dfrac{d^2y}{dx^2}+y=3\sin x$

3. (1) $\oint_C\left(\dfrac{a^2y^2}{b^2}+\dfrac{b^2x^2}{a^2}\right)^{\frac{1}{2}}ds$ を求めよ．ただし，C は $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$ の楕円とする．

(2) 球面 $x^2+y^2+z^2=a^2$ と円柱面 $x^2+y^2=ax$ で囲まれた立体の体積 V を求めよ（$a>0$）．

問題 II

1. (1) 3 次元のベクトル $\boldsymbol{a} = (a_x, a_y, a_z)$, $\boldsymbol{b} = (b_x, b_y, b_z)$, $\boldsymbol{c} = (c_x, c_y, c_z)$ について次のベクトル式が成り立つ.

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = (\boldsymbol{a} \cdot \boldsymbol{c})\,\boldsymbol{b} - (\boldsymbol{a} \cdot \boldsymbol{b})\,\boldsymbol{c}$$

$\boldsymbol{b} \times \boldsymbol{c} = \boldsymbol{u} = (u_x, u_y, u_z)$ とおき，各成分を計算することにより上式が成り立つことを確かめよ.

(2) スカラー3 重積を $[\boldsymbol{a}\,\boldsymbol{b}\,\boldsymbol{c}] = (\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{c}$ と表すとき，上の結果を用いて次式が成り立つことを示せ.

$$(\boldsymbol{a} \times \boldsymbol{b}) \times (\boldsymbol{c} \times \boldsymbol{d}) = [\boldsymbol{a}\,\boldsymbol{b}\,\boldsymbol{d}]\,\boldsymbol{c} - [\boldsymbol{a}\,\boldsymbol{b}\,\boldsymbol{c}]\,\boldsymbol{d}$$

2. (1) 次の実対称行列 A を直交行列 P によって対角化せよ.

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

(2) 上の結果を用いて，次の二次形式を直交変換を用いて標準形（$\lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$ の形）にせよ.

$$f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 4x_3^2 + 2x_1 x_2$$

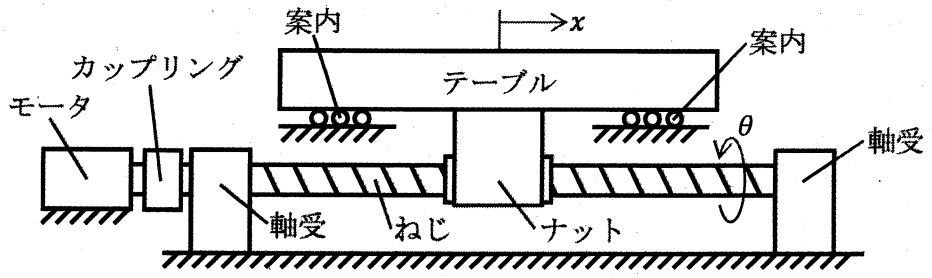
(3) $x_1^2 + x_2^2 + x_3^2 = 1$（ただし，$x_1, x_2, x_3$ は実数）とするとき，2.(2)で定義した二次形式について次式が成り立つことを示せ.

$$1 \leq f(x_1, x_2, x_3) \leq 4$$

問題 III

下図に示す送りシステムについて考える．モータのトルクTはカップリングを介してねじに伝えられ，その回転運動はナットに接続されたテーブルの並進運動に変換される．回転部分は回転軸周りの慣性モーメントJを有し，並進部分は質量mを有する．回転角度をθ[rad]とし，並進位置をxとする．xの正の方向は，θが増える時にxが増えるように定める．ねじとナットのリードはlである．ねじは２箇所の軸受で支持されて回転方向のみに運動し，テーブルは案内されて並進方向のみに運動する．摩擦力，弾性変形，および空気抵抗は無視することができる．

1. 送りシステムとその運動について，以下の問いに答えよ．
   (1) ねじとナットの間で，角度θは位置xに変換され，並進部分を駆動するトルクT_nは力F_nに変換される．θとx，T_nとF_nの関係式をそれぞれ求めよ．
   (2) TおよびT_nを用いて回転部分の回転の運動方程式を求めよ．
   (3) F_nを用いて並進部分の並進の運動方程式を求めよ．
   (4) モータトルクTと角加速度$\ddot{\theta}$の関係が$T = J_e\ddot{\theta}$によって表されることを示し，定数J_eを求めよ．
2. 初期の並進速度と位置をゼロ，すなわち時間$t = 0$において$\dot{x} = 0$，$x = 0$とする．モータが，$0 \le t < t_1$の間トルクT_1を発生し，その後$t_1 \le t < 2t_1$の間トルク$-T_1$を発生する．J_eを用いて以下の問いに答えよ．
   (1) 横軸を時間t，縦軸を並進加速度$\ddot{x}$として両者の関係を表すグラフを描き，グラフ内に時間$\frac{1}{2}t_1$，$\frac{3}{2}t_1$における加速度，および時間$0, t_1, 2t_1$をそれぞれ記入せよ．
   (2) 横軸を時間t，縦軸を並進速度$\dot{x}$として両者の関係を表すグラフを描き，グラフ内に時間t_1における速度を記入せよ．
   (3) 横軸を時間t，縦軸を並進位置xとして両者の関係を表すグラフを描き，グラフ内に時間t_1および$2t_1$における位置をそれぞれ記入せよ．



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Specialized Subjects

Number of pages (including this cover sheet): 5

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Notes

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Problem I

1. Calculate the following formulae.

(1) $\dfrac{d}{dx}\left(x^{x\sin x}\right)$

(2) $\lim\limits_{x\to 0}\dfrac{1-\cos x}{x\sin x}$

2. Find the general solutions of the following differential equations.

(1) $e^{2x-y}+e^{x+y}\dfrac{dy}{dx}=0$

(2) $\dfrac{d^2y}{dx^2}+y=3\sin x$

3. (1) Calculate $\oint_C\left(\dfrac{a^2y^2}{b^2}+\dfrac{b^2x^2}{a^2}\right)^{\frac{1}{2}}ds$. Here, let C denote the ellipse $\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$.

(2) Find the volume V of the solid bounded by the spherical surface $x^2+y^2+z^2=a^2$ and the cylindrical surface $x^2+y^2=ax$, where $a>0$.

Problem II

1. (1) For three-dimensional vectors $\boldsymbol{a} = (a_x, a_y, a_z)$, $\boldsymbol{b} = (b_x, b_y, b_z)$, $\boldsymbol{c} = (c_x, c_y, c_z)$, the following vector relation holds:

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = (\boldsymbol{a} \cdot \boldsymbol{c})\,\boldsymbol{b} - (\boldsymbol{a} \cdot \boldsymbol{b})\,\boldsymbol{c}$$

Let $\boldsymbol{b} \times \boldsymbol{c} = \boldsymbol{u} = (u_x, u_y, u_z)$, and verify this relation by calculating each component of the equation.

(2) Let $[\boldsymbol{a}\,\boldsymbol{b}\,\boldsymbol{c}] = (\boldsymbol{a} \times \boldsymbol{b}) \cdot \boldsymbol{c}$ denote the scalar triple product. Using the result above, prove the following relation:

$$(\boldsymbol{a} \times \boldsymbol{b}) \times (\boldsymbol{c} \times \boldsymbol{d}) = [\boldsymbol{a}\,\boldsymbol{b}\,\boldsymbol{d}]\,\boldsymbol{c} - [\boldsymbol{a}\,\boldsymbol{b}\,\boldsymbol{c}]\,\boldsymbol{d}$$

2. (1) Diagonalize the following real symmetric matrix A by an orthogonal matrix P:

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

(2) Using the result obtained above, express the following quadratic form in its standard form ($\lambda_1 y_1^2 + \lambda_2 y_2^2 + \lambda_3 y_3^2$) by an orthogonal transformation:

$$f(x_1, x_2, x_3) = 2x_1^2 + 2x_2^2 + 4x_3^2 + 2x_1 x_2$$

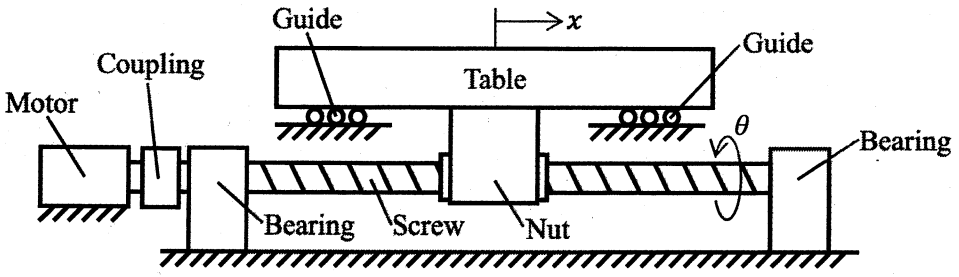
(3) Let $x_1^2 + x_2^2 + x_3^2 = 1$ (where x_1, x_2, x_3 are real numbers). Prove that, for the quadratic form defined in 2.(2), the following inequality holds:

$$1 \leq f(x_1, x_2, x_3) \leq 4$$

Problem III

Consider the feed system shown in the following figure. Torque T of the motor is transmitted to the screw via the coupling, and its rotational motion is converted to translational motion of the table connected to the nut. The rotational part has moment of inertia J around the rotation axis, while the translational part has mass m. Let the rotational angle be θ[rad] and the translational position be x. The positive direction of x is determined so that x increases as θ increases. Lead of the screw and the nut is l. The screw is supported by the two bearings to move only in the rotational direction, and the table is guided to move only in the translational direction. Frictional force, elastic deformation, and air resistance are negligible.

1. Answer the following questions about the feed system and its motion.
   (1) Between the screw and the nut, the angle θ is converted to the position x, and the torque T_n is converted to the force F_n to drive the translational part. Derive the equations relating θ and x, and T_n and F_n, respectively.
   (2) Derive the rotational equation of motion of the rotational part, utilizing T and T_n.
   (3) Derive the translational equation of motion of the translational part, utilizing F_n.
   (4) Demonstrate that relation between the motor torque T and the angular acceleration $\ddot{\theta}$ can be expressed by $T = J_e\ddot{\theta}$, and find the constant J_e.
2. Suppose the initial translational velocity and position to be zero, i.e., $\dot{x} = 0$ and $x = 0$, respectively, at time $t = 0$. The motor generates torque T_1 during a period of $0 \leq t < t_1$ and then torque $-T_1$ during a period of $t_1 \leq t < 2t_1$. Answer the following questions, utilizing J_e.
   (1) Plot a graph relating the time t and the translational acceleration $\ddot{x}$ by setting t on the horizontal axis and $\ddot{x}$ on the vertical axis. Write the accelerations at the times $\frac{1}{2}t_1$ and $\frac{3}{2}t_1$ and the times 0, t_1, and $2t_1$ on the graph.
   (2) Plot a graph relating the time t and the translational velocity $\dot{x}$ by setting t on the horizontal axis and $\dot{x}$ on the vertical axis. Write velocity at time t_1 on the graph.
   (3) Plot a graph relating the time t and the translational position x by setting t on the horizontal axis and x on the vertical axis. Write positions at times t_1 and $2t_1$ on the graph.


Guide
x
Guide
Coupling
Table
Motor
θ
Bearing
Bearing
Screw
Nut